MyLearna
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MATHEMATICS

Name: ____________________

Date: ____________________

Use number relationships to support algebraic thinking

Step 50

Look for patterns and generalize relationships.

Today I am learning to use number relationships to support algebraic thinking.

1. Find the Pattern

Look at each pattern. Write the next three terms.

a) 2, 5, 8, 11, 14, ...

_____, _____, _____

What is the rule?

____

b) 3, 6, 9, 12, 15, ...

_____, _____, _____

What is the rule?

____

c) 5, 10, 15, 20, 25, ...

_____, _____, _____

What is the rule?

____

d) 1, 4, 7, 10, 13, ...

_____, _____, _____

What is the rule?

____

Think about it!

- How do the numbers change each time?
- Can you describe the pattern in words?
- How can the rule help you find any term?

2. Complete the Table

Use the given rule to complete each table.

a) Rule: add 4

Input (n)	1	2	3	4	5
Output	5				

b) Rule: multiply by 3

Input (n)	1	2	3	4	5
Output	3				

c) Rule: subtract 6

Input (n)	6	7	8	9	10
Output	0				

d) Rule: double, then add 1

Input (n)	1	2	3	4	5
Output	3				

Remember:

- A rule shows how the output numbers depend on the input numbers.
- You can write a rule in words.
- The rule can help you find any missing term.

3. Write a Rule

Write a rule for each pattern.

a) 6, 11, 16, 21, 26, ...

Rule: ____________________

b) 100, 90, 80, 70, 60, ...

Rule: ____________________

c) 7, 14, 21, 28, 35, ...

Rule: ____________________

d) 2, 5, 8, 11, 14, ...

Rule: ____________________

Think about it!

- What operation is being done?
- What number is the operation being done to?
- Test your rule with one of the terms.

4. Use the Rule

Use the rule to find the missing term or a specific term.

a) Rule: add 7
24, 31, 38, 45, _____, _____
What is the 10th term? __________

b) Rule: subtract 3
50, 47, 44, 41, _____, _____
What is the 15th term? __________

c) Rule: multiply by 4
4, 8, 12, 16, _____, _____
What is the 20th term? __________

d) Rule: double, then subtract 2
0, 2, 6, 10, _____, _____
What is the 25th term? __________

Remember:

- Use the rule repeatedly or use multiplication to jump ahead.
- Check your answer by continuing the pattern.

5. Generalize

Write an expression (rule) for the nth term.

a) 3, 7, 11, 15, 19, ...

nth term: ____________________

b) 2, 6, 10, 14, 18, ...

nth term: ____________________

c) n, 2n, 3n, 4n, 5n, ...

nth term: ____________________

d) 5, 10, 15, 20, 25, ...

nth term: ____________________

Think about it!

- Use *n* to represent the term number (1st term is $n = 1$, 2nd term is $n = 2$, ...).
- Look for what is being multiplied by *n*, or what is being added/subtracted.

6. Problem Solving

Solve each problem using a number relationship.

a) A pattern starts at 4. Each term is 5 more than the previous term. What is the 30th term?

Rule: ____________________

Answer: ____________________

b) A pattern starts at 120. Each term is 12 less than the previous term. What is the 25th term?

Rule: ____________________

Answer: ____________________

c) A pattern is made by multiplying the term number by 7. What is the 18th term?

Rule: ____________________

Answer: ____________________

d) A pattern is made by doubling the term number and adding 3. What is the 40th term?

Rule: ____________________

Answer: ____________________

Think about it!

- Write the rule first.
- Decide if you need to add, subtract, multiply or divide.
- Use the rule to find the required term.

My Learning Check

Circle how confident you feel.

I can do it! | Almost there! | I need more practice. | I need help.

Parent Observation (optional)

Please tick the boxes that apply.

- ☐ Understands number patterns
- ☐ Finds rules from patterns
- ☐ Uses rules to find terms
- ☐ Writes expressions for nth term
- ☐ Solves problems using rules
- ☐ Explains thinking clearly
- ☐ Needs more practice

Notes: ____________________

Today I learned...

1 thing I did well: ____________________

1 thing I will improve: ____________________